Auditable repair of identity leakage in group-structured data

feasibility decided before any dose is chosen

1 THE PROBLEM



MODEL

in-sample: excellent

leave-one-group-out collapses

recognised the group, not the target

2 THE AUDITABLE LOOP

DETECT LDI under leave-one-group-out

PRE-COMPUTE signal-to-null ratio vs 2.35

REPAIR adversarial training, dose λ

RE-AUDIT safety, leakage, non-blindness

Boundary I coupling
decides repairability

Boundary II group count
decides signal survival

ACCEPT
signal kept, leakage below threshold

REJECT
no feasible point, predicted in advance

3 WHERE EACH CLAIM IS VALIDATED

VALIDATED end-to-end: repair, acceptance, endpoint
spatial transcriptomics, 3 cohorts + simulator

AUDITED instrument and boundaries on real data, no endpoint
wearables, chemical sensing

NOT TESTED structural mapping only, no experiment
imaging, speech, EHR, remote sensing, trials